

BRAZESH GURAGAIN

Full Stack Developer & AI Integration Engineer

Hobart, Tasmania, Australia | [Linkedin](#) | [Github](#) | brazeshguragain.com | b.guragain1023@gmail.com

PROFESSIONAL SUMMARY

Full Stack Developer with a Master of Technology (Software Engineering) and hands-on experience building production-grade web applications and AI-powered systems using React, Node.js, Express, MongoDB, and LLM APIs. Shipped three end-to-end deployed projects including a live RAG application with vector search, embedding pipelines, and LLM generation — built independently from concept to production. Strong communication and problem-solving skills developed across 3+ years of customer-facing roles.

TECHNICAL SKILLS

Frontend: HTML5, CSS3, JavaScript (ES6+), React, Bootstrap 5, react-markdown, Responsive Design

Backend: Node.js, Express.js, MongoDB, Mongoose, REST APIs, Middleware

AI & Vector: Anthropic Claude API, OpenAI API (Embeddings + GPT-5 Nano), RAG Architecture, LanceDB, Prompt Engineering, Semantic Search, Cosine Similarity, Text Chunking

Tools & Deployment: Git, GitHub, VS Code, Vercel, Railway, Render

PROJECTS

Aussie in WorldCup 2026 — Live RAG Chatbot

Full-stack RAG application providing accurate, context-aware answers about the Australian Socceroos at the 2026 FIFA World Cup

- Built a complete RAG pipeline from scratch: paragraph-based sliding window chunking → OpenAI text-embedding-3-small embeddings → LanceDB vector storage → cosine similarity retrieval → Claude API generation
- Implemented a two-stage retrieval pipeline: semantic search retrieves top-K chunks, Claude API re-ranks results before final answer generation — reducing hallucinations and improving answer accuracy
- Engineered LanceDB integration including custom deletion logic (drop-and-recreate table pattern) to maintain sync between MongoDB document store and vector database across all CRUD operations
- Built a password-protected admin dashboard with backend-validated auth, document upload with automatic chunking and ingestion, and synced deletion across both MongoDB and LanceDB
- Deployed frontend to Vercel and backend to Railway with persistent volume for LanceDB data — configured dynamic path resolution between local development and production environments
- Tech stack: React, Vite, Bootstrap 5, react-markdown (frontend) | Node.js, Express, MongoDB/Mongoose, LanceDB, OpenAI API, Anthropic Claude API (backend)

AI Prompt Diff — LLM Comparison Tool

Side-by-side LLM response comparison tool with AI-generated diff analysis

- Sends a single prompt to Claude Sonnet and GPT-5 Nano simultaneously using parallel async API calls, comparing responses side by side with metrics including length, clarity score, and response speed
- AI-generated summary of the 3 key differences between both responses — discovered GPT-5 Nano used 26x more tokens than Claude for the same prompt due to internal reasoning chain
- Built with React, Node.js, Express, Anthropic API, OpenAI API — deployed on Vercel with serverless functions

Time Tracker — Full Stack Task Management App

Full-stack productivity app for tracking weekly hours and identifying inefficiencies

- Built with React frontend and Node.js/Express backend with MongoDB Atlas for cloud data persistence
- Bulk selection with Select All checkbox — delete single, multiple or all items in one action with real-time PATCH calls
- Deployed backend on Render with UptimeRobot monitoring to prevent free-tier spin-down

CineTrial — Movie Discovery App

React app powered by the TMDB API with watchlist persistence

- Fetches a random movie on load for instant discovery; browse trending, drama and comedy with one-click filtering
- Save movies to a personal watchlist that persists across sessions using local storage

EDUCATION

Master of Technology — Software Engineering | Federation University Australia | 2019

IIBIT Sydney Campus — Postgraduate qualification covering software design, systems architecture and engineering principles

Full Stack Development Bootcamp | Dented Code Academy | Jan 2026 – Present

Completed MERN stack and AI integration — building real-world projects end to end

WORK EXPERIENCE

Support Worker | Christian Homes Tasmania | 2023 – Present | Hobart, Tasmania

- Provided tailored support to individuals with diverse needs, adapting communication style for each client
- Maintained accurate case notes and collaborated with multidisciplinary teams to deliver care plans
- Demonstrated empathy, patience and reliability in high-pressure and emotionally demanding environments

Team Member | Woolworths | 2022 – Present | Hobart, Tasmania

- Delivered consistent customer service in a fast-paced retail environment serving hundreds of customers daily
- Managed inventory, restocking and point-of-sale operations efficiently